



AUDITVECTOR SYSTEM ARCHITECTURE

AUTONOMOUS FINANCIAL INTEGRITY INVESTIGATOR & EVIDENCE-GROUNDED AUDIT ENGINE

Google ADK 2.7

Gemini 3.5 Flash

Google Cloud Run

Cloud Pub/Sub

Cloud Firestore

DuckDB SQL

01 / CLIENT & INGESTION

PORT 8000

Forensic Command Console

Single-page interactive dark UI featuring 3-column live multi-agent command center, Claim vs. Reality comparators, and slide-out inspector.

HTML5 / Vanilla JS / CSS3

FastAPI REST Service

Asynchronous API managing audit lifecycles, demo triggers (/demo/alpha, /demo/control, /demo/aibip), and evidence bundle exports.

FastAPI • Uvicorn • Pydantic

Multi-Source Adapters

Ingests raw trade logs (CSV with SHA-256 anchors), strategy configs (JSON), and performance summaries.

CSVIngestionAdapter • ReportAdapter

Secret Redactor

Pre-execution sanitization stripping API keys, Bearer tokens, and RSA private keys before LLM processing.

SecretRedactor (Regex Engine)

02 / CLOUD RUNTIME

ASYNC WORKER

Google Cloud Run

Serverless container runtime hosting the FastAPI backend and orchestrator microservices.

deploy_cloudrun.sh • Dockerfile

Google Cloud Pub/Sub

Decoupled asynchronous event broker (Topic: auditvector-jobs) dispatching work to background workers.

GCPPubSubPublisher • Push Webhook

Idempotent Worker

Stage-aware worker preventing duplicate execution of active or completed audits upon message replay.

PubSubAuditWorker • State Sync

Google Cloud Firestore

Distributed persistent document store synchronizing real-time audit stages (CREATED → QUEUED → COMPLETED).

FirestoreAuditStateStore

03 / 5-AGENT ADK PIPELINE

TASKMASTER

01 AuditPlanner

Scopes repository structure, discovers transactional datasets, and formulates 9-step audit strategy.

02 RepositoryInvestigator

Parses Python AST syntax trees to map financial routines (PnL, fees, returns) without dumping entire codebases.

03 FinancialInvestigator

Extracts claimed performance metrics and config parameters into typed FinancialClaim targets.

04 ContradictionInvestigator

Dispatches deterministic verifier tools, calculates mathematical variances, and assigns FindingStatus.

05 ReportAgent

Aggregates verified findings, tallies capital at risk, and renders executive Markdown & provenance graphs.

04 / DETERMINISTIC ENGINE

0 HALLUCINATIONS

FIFO Lot Matching Engine

Deterministic bottom-up reconstruction of realized PnL for Long & Short inventory queues from raw execution fills.

PnLRecalculator v2.2 (Python)

DuckDB Tabular Analytics

High-throughput in-memory analytical SQL execution profiling trade distributions and volume metrics.

DuckDB 1.5.5 In-Memory

Fee & Polarity Verifiers

Automated detectors for fee double-counting, model rate drift, and disguised negative-to-positive return inversions.

FeeRecalculator • ReturnCalculator

Cryptographic Evidence Contracts

Strict gate: No Evidence Contract → No Verified Finding. Links source line citations, SHA-256 hashes, and verifier math.

EvidenceContract • EvidenceGraph

Core Design Mandate:

"AI reasons. Code proves. Evidence explains."

• 100% Deterministic Arithmetic Grounding

All Things Agentic Hackathon • Track: The Taskmaster • Paramjeet Singh